

SE LAB 3 JUSTIFICATION

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Architectural Style Analysis

1) Layered Architecture

Layered Architecture organizes the system into horizontal layers, where each layer has a specific responsibility. For the Doctor Consultation and Booking System, the layers can be divided into a Presentation Layer, Business Layer, and Data Layer. The Presentation Layer handles the patient and doctor interface, the Business Layer manages functions such as doctor availability, appointment booking, online consultation, and prescription management, while the Data Layer stores information in the database.

Advantages: It provides a clear separation of concerns, making the system easier to understand, develop, test, and maintain. Changes made to one layer can generally be managed without directly modifying the responsibilities of other layers.

Disadvantages: Communication between multiple layers can introduce performance overhead. There can also be tight coupling between layers, and scaling individual components independently can be difficult.

Suitability: This architecture is suitable for the system because the major functions can be clearly separated into presentation, business logic, and data management.

2) Microservice Architecture

Microservices Architecture divides the application into small, independently deployable services. In the Doctor Consultation and Booking System, separate services could handle doctor discovery, appointment booking, online consultation, prescription management, and doctor availability.

Advantages: Individual services can be scaled independently, and failures can potentially be isolated to a particular service. Different technologies can also be used for different services when required.

Disadvantages: Microservices introduce operational complexity because multiple independent services must communicate over a network. They can also introduce network latency and create challenges related to maintaining data consistency between services.

Suitability: This architecture could be useful for a large-scale healthcare platform with many users and independently evolving services. However, for the current system, the additional operational complexity may not be necessary.

3) Client-Server Architecture

Client-Server Architecture consists of clients communicating with a centralized server. In this system, the patient and doctor interfaces would act as clients, while the server would handle appointment booking, doctor information, consultations, prescriptions, and database operations.

Advantages: It provides centralized control and makes deployment relatively simple. Data can also be maintained centrally, which helps maintain consistency.

Disadvantages: The centralized server can become a single point of failure and may become a scalability bottleneck as the number of users increases. It also provides less fault tolerance compared with architectures where services are distributed.

Suitability: Client-Server architecture can support the basic requirements of the system, but it provides less separation between the different business functionalities.

Selected Architecture: Layered Architecture

For the Online Doctor Consultation and Booking System, Layered Architecture was selected. The system is divided into three main layers: Presentation Layer, Business Layer, and Data Layer. The Presentation Layer contains the Patient/Doctor UI, the Business Layer contains the core system components, and the Data Layer contains the Database.

Reason 1 – Separation of Responsibilities

Layered Architecture provides a clear separation of responsibilities between different parts of the system. The Presentation Layer handles interaction with patients and doctors, while the Business layer handles functions such as doctor and availability management, appointment booking, online consultations, and prescription management. The Data Layer is responsible for storing and retrieving system data. This separation makes the system easier to understand, develop, test, and maintain.

Reason 2 – Easier Maintenance and Modification

The layered structure allows changes to one part of the system to be made with less impact on the other layers. For example, changes to the user interface can be made without directly modifying the database. Similarly, business functionality such as appointment booking or prescription management can be modified while keeping the presentation layer separate. This makes the system easier to maintain and extend as new features are added.

Security Advantage

Layered Architecture improves security by preventing the user interface from directly accessing the database. Requests from the Patient/Doctor UI pass through the business components before accessing stored data. This allows access to sensitive information such as patient, consultation, and prescription data to be controlled through the business layer. The separation of the Data Layer also reduces direct exposure of the database to users.

Performance Benefit

The architecture separates frequently used operations into individual business components, such as doctor availability, appointment booking, and online consultation. This allows each operation to process only the data and functionality required for that request. Database access can also be handled through the business layer, avoiding unnecessary direct communication between the user interface and database.